Roll out the personalized roaming prompt to unlock CHF 4.7M annual margin with no guardrail breach.

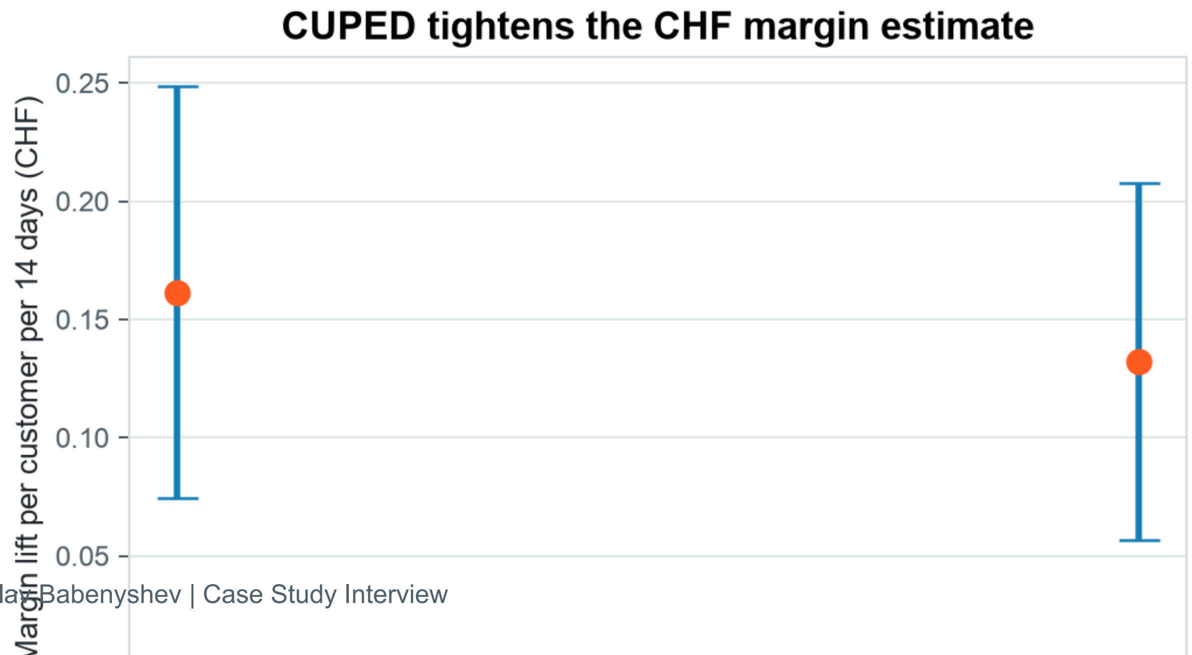
- A customer-level randomized A/B test on 60,000 synthetic Sunrise app customers shows a conversion lift of **1.10 pp**.
- CUPED estimates **CHF 0.13** incremental margin per eligible customer per 14 days, with a 95% CI of **CHF 0.06 to CHF 0.21**.
- Recommendation: launch through a monitored 25% ramp, then scale to all eligible app customers if guardrails remain green.

The business request was translated into a randomized revenue-and-experience decision framework.

Design choice	Specification
Stakeholder decision	Ship, iterate, or stop a personalized roaming-pack prompt
Randomization unit	Customer, assigned 50/50 to control or treatment
Target population	Eligible Sunrise mobile-app customers before summer travel
Primary metric	14-day roaming-pack conversion rate
Business metric	14-day gross margin CHF per customer
Decision rule	Significant conversion and CHF lift, with all guardrails passing

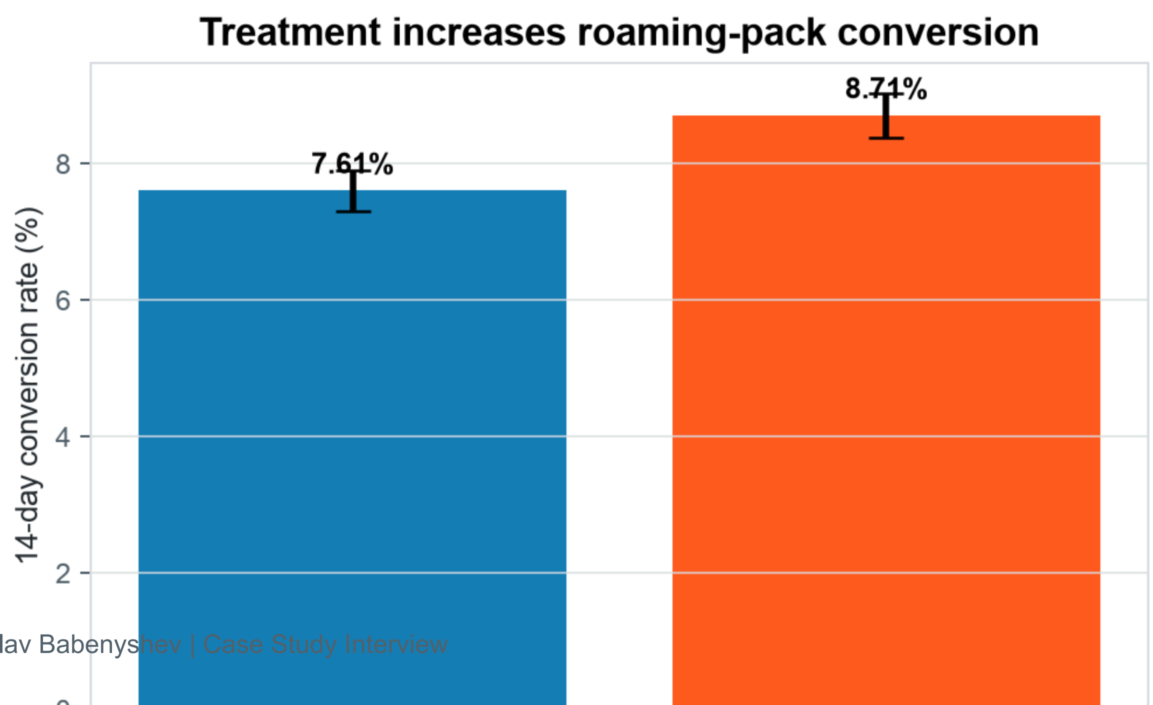
CUPED is the right method because pre-period roaming spend explains margin variation without changing randomization.

- Standard conversion test: two-sided difference in proportions at alpha = 5%.
- CHF estimate: Welch mean difference on margin, plus CUPED using pre-period roaming revenue.
- Rigor: customer-level randomization avoids pre/post bias; balance checks reduce sample-selection risk; guardrails prevent revenue-only launch logic.



The treatment creates CHF 4.7M expected annual margin versus the current app experience.

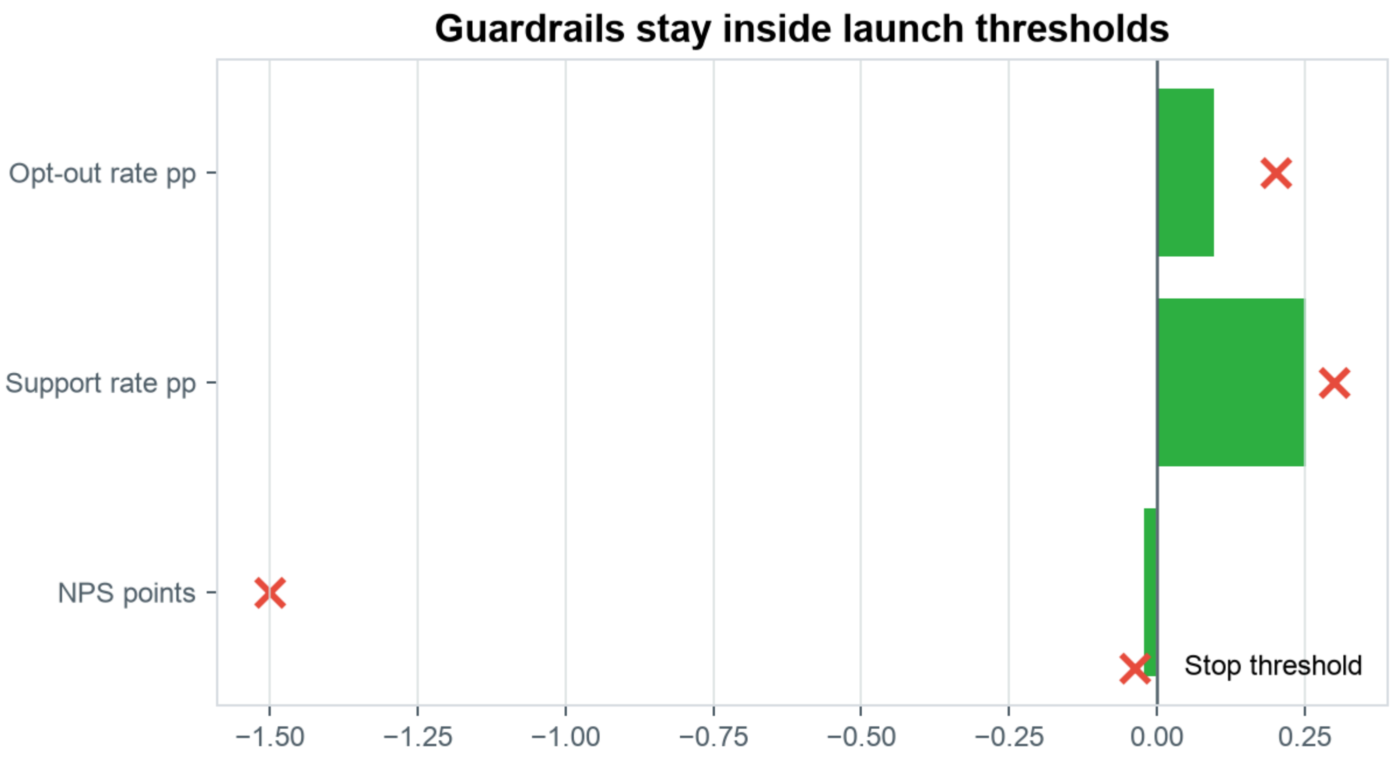
Scenario	Conversion	Margin lift	Annualized margin
Current app experience	7.61%	CHF 0.00	CHF 0.0M
Personalized prompt	8.71%	CHF 0.13 / customer / 14d	CHF 4.7M
Conservative CI floor	+0.66%	CHF 0.06 / customer / 14d	CHF 2.0M



A one-week ramp is sufficient to validate production performance before full scale.

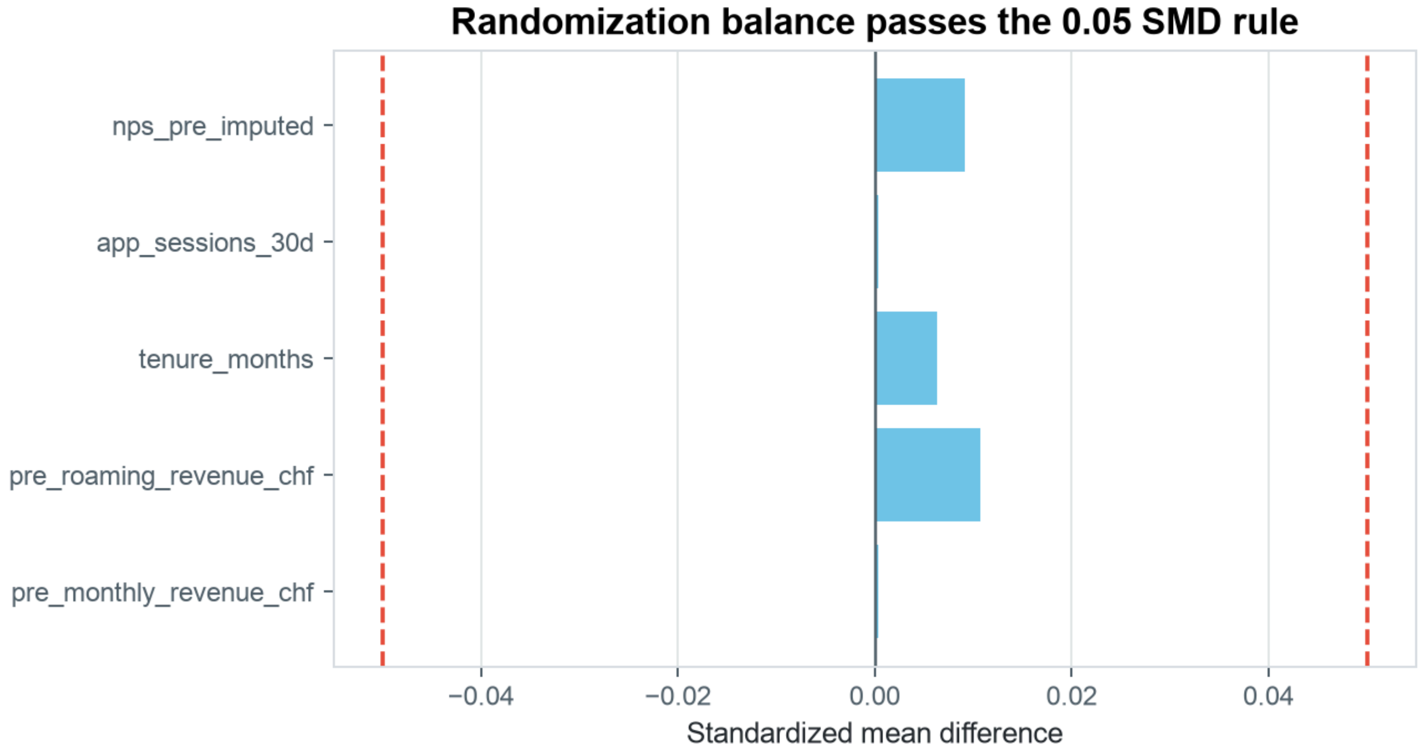
Launch control	Requirement
Power design	30,655 customers per group to detect a 0.6 pp MDE
Traffic assumption	8,500 eligible customers per day
Minimum test duration	8 days for a fully powered confirmatory read
Ramp proposal	25% for one week, 50% for one week, then 100% if guardrails pass
Guardrails	NPS, support contacts, and app opt-outs monitored daily

Customer-experience guardrails stayed green, so the next decision is operational rollout approval.



- NPS difference stayed inside the -1.5 point stop threshold.
- Support contact and app opt-out differences stayed below their rate thresholds.
- Retain a 5% post-launch holdout for four weeks to measure persistence and seasonality.

Appendix: Randomization balance supports a causal read of the treatment effect.



- All audited pre-treatment covariates are inside the absolute SMD < 0.05 balance rule.
- Balance was checked before outcome analysis to avoid tuning the design after seeing results.
- Synthetic source tables mimic CRM, billing, app engagement, assignment, and outcome systems.

Appendix: CUPED variance reduction increases precision without changing the estimand.

Diagnostic	Result
CUPED theta	0.245
Standard margin lift	CHF 0.16
CUPED margin lift	CHF 0.13
Variance reduction	25.2%
CUPED p-value	5.85e-04

- Pre-period roaming revenue is measured before randomization, so it improves precision without absorbing treatment impact.

Appendix: The project artifacts provide a reproducible audit trail for review and rerun.

```
graph LR
    A[Raw mock source tables] --> B[Joined analysis table]
    B --> C[Balance and missingness checks]
    B --> D[Conversion test]
    B --> E[CUPED margin test]
    B --> F[Guardrail tests]
    C --> G[Notebook audit trail]
    D --> H[Executive deck]
    E --> H
    F --> H
```

- Notebook: `notebooks/ab-test-mock-data.ipynb`
- Source and processed artifacts: `notebooks/ab-test-mock-data/`
- Stakeholder deck: `docs/ab-test-mock-data/deck.md`